

Assignment: End-to-End Exploratory Data Analysis (EDA), Data Cleaning, Outlier Detection, Feature Engineering, and Machine Learning Readiness

Dataset: Hotel Booking Demand Dataset

Business Scenario

You have been hired as a Data Analyst by a multinational hotel chain. The management wants to understand booking behavior, cancellation patterns, customer preferences, seasonal trends, revenue opportunities, and operational inefficiencies.

Your objective is to perform a complete End-to-End EDA, Data Cleaning, Data Quality Assessment, Feature Engineering, and Machine Learning Readiness Analysis to support business decision-making.

Part 1: Dataset Understanding and Initial Exploration

Task 1.1: Dataset Loading

Perform:

1. Load the dataset.
2. Display first 10 records.
3. Display last 10 records.
4. Check shape of dataset.
5. Display all columns.
6. Identify numerical and categorical columns.
7. Check memory usage.

Questions

1. How many bookings are present?
 2. How many features are available?
 3. How many numerical features exist?
 4. How many categorical features exist?
 5. Which columns appear to be identifiers?
 6. Which columns seem useful for predicting cancellations?
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Task 1.2: Dataset Overview

Generate:

- `info()`
- `describe()`
- `describe(include='object')`
- data type summary

Questions

1. Which feature has the highest average value?
 2. Which feature has the largest variance?
 3. Which feature has the smallest variance?
 4. Which feature has the highest cardinality?
 5. Which feature contains the most distinct categories?
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Part 2: Data Quality Assessment

Task 2.1: Missing Values Analysis

Calculate:

- Total Missing Values
- Missing Value Percentage

Create:

- Missing Value Table

- Missing Value Heatmap
- Missing Value Bar Chart

Business Insight Questions

1. Which columns have missing values?
 2. Which column has the highest percentage of missing values?
 3. Why might guests not provide this information?
 4. Would removing these records cause information loss?
 5. Which columns should be imputed and which should be dropped?
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Task 2.2: Duplicate Analysis

Perform:

1. Detect duplicates.
2. Count duplicates.
3. Remove duplicates.

Questions

1. How many duplicate bookings exist?
 2. Why can duplicate records distort hotel analytics?
 3. How would duplicates impact revenue forecasting?
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Task 2.3: Unique Value Analysis

For every feature:

Calculate:

- Number of unique values
- Most frequent value
- Frequency percentage

Questions

1. Which feature has the highest cardinality?
2. Which feature may act as a grouping variable?

3. Which feature would be difficult to encode directly?

Part 3: Data Cleaning

Task 3.1: Missing Value Treatment

For Numerical Columns:

Apply:

- Mean Imputation
- Median Imputation

For Categorical Columns:

Apply:

- Mode Imputation
- Unknown Category Imputation

Questions

1. Which strategy worked best?
 2. Why is median preferred for skewed data?
 3. Which columns required special treatment?
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Task 3.2: Data Type Validation

Validate:

- Numerical Columns
- Categorical Columns
- Date Columns

Questions

1. Were any incorrect data types found?
2. How could incorrect data types affect business reports?
3. Which columns should be converted into categories?

Part 4: Univariate Analysis

Task 4.1: Numerical Feature Analysis

For Every Numerical Feature

Calculate:

- Mean
- Median
- Mode
- Variance
- Standard Deviation
- Min
- Max
- Range
- Quartiles
- Skewness

Create:

- Histogram
- KDE Plot
- Boxplot

Business Insight Questions

1. What is the average lead time of bookings?
2. How far in advance do guests usually book?
3. Which feature has the highest variability?
4. Are there extreme booking behaviors?
5. Which variables are highly skewed?

Task 4.2: Categorical Feature Analysis

For Every Categorical Feature

Calculate:

- Frequency Count
- Percentage Distribution

Create:

- Countplots
- Pie Charts

Business Insight Questions

1. Which hotel receives more bookings?
 2. Which market segment dominates bookings?
 3. Which customer type is most common?
 4. Which distribution channel generates most bookings?
 5. Which country contributes most guests?
 6. Which meal package is most popular?
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Part 5: Hotel Business Analysis

Task 5.1: Cancellation Analysis

Analyze:

- is_canceled

Create:

- Countplot
- Pie Chart

Important Business Questions

1. What is the cancellation rate?
 2. What percentage of bookings are successfully completed?
 3. How much potential revenue may be lost due to cancellations?
 4. Which hotel experiences higher cancellations?
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Task 5.2: Seasonal Booking Analysis

Create:

- Monthly booking trend
- Year-wise booking trend
- Month-Year heatmap

Questions

1. Which month receives maximum bookings?
 2. Which month receives minimum bookings?
 3. Is there seasonality in hotel demand?
 4. During which season should staffing be increased?
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Part 6: Outlier Detection

Task 6.1: IQR Method

For every numerical feature:

Calculate:

- Q1
- Q3
- IQR
- Lower Bound
- Upper Bound

Identify:

- Number of Outliers

Create:

- Boxplots

Questions

1. Which feature contains most outliers?
 2. Are outliers genuine customer behavior or data errors?
 3. Which outliers represent high-value bookings?
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Task 6.2: Z-Score Method

Calculate:

- Z-score

Identify:

$$|Z| > 3$$

Questions

1. Compare IQR and Z-score methods.
 2. Which method works better on skewed hotel data?
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Part 7: Outlier Treatment

Apply:

Method 1

Remove Outliers

Method 2

Capping

Method 3

Winsorization

Questions

1. Which method preserves maximum data?
 2. Which method changes distribution significantly?
 3. Which method would be recommended for business analytics?
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Part 8: Feature Engineering

Create At Least 10 New Features

Feature 1

Total Guests

Adults + Children + Babies

Feature 2

Stay Duration

Weekend Nights + Week Nights

Feature 3

Booking Season

Winter / Summer / Monsoon

Feature 4

Lead Time Category

Short / Medium / Long

Feature 5

Guest Type

Solo / Couple / Family

Feature 6

Booking Risk Score

Based on:

- Lead Time
- Previous Cancellations

- Deposit Type

Feature 7

Revenue Opportunity Score

Based on:

- Stay Duration
- ADR

Feature 8

High Value Customer

Based on ADR percentile

Feature 9

Advance Booking Group

Feature 10

Special Request Category

Low / Medium / High

Questions

1. Which engineered feature is most useful?
2. Which feature can help predict cancellations?
3. Which feature helps revenue analysis most?

Part 9: Bivariate Analysis

Numerical vs Numerical

Create:

- Correlation Matrix

- Heatmap
- Scatterplots

Questions

1. Which variables are highly correlated?
 2. Does lead time affect ADR?
 3. Does stay duration affect revenue?
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Numerical vs Categorical

Create:

- Boxplots
- Violinplots
- Grouped Statistics

Questions

1. Which hotel has the highest ADR?
 2. Which market segment spends most?
 3. Which customer type stays longest?
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Categorical vs Categorical

Create:

- Crosstabs
- Heatmaps

Questions

1. Is cancellation associated with deposit type?
 2. Is cancellation associated with market segment?
 3. Is cancellation associated with customer type?
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Part 10: Deep Business Insight Analysis

Revenue Analysis

Create:

Estimated Revenue

ADR × Stay Duration

Questions

1. Which hotel generates highest revenue?
 2. Which month generates highest revenue?
 3. Which customer segment generates most revenue?
 4. Which country contributes most revenue?
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Cancellation Risk Analysis

Questions

1. Which customer type cancels most?
 2. Which market segment cancels most?
 3. Which booking channel has highest cancellation?
 4. Which deposit type has highest cancellation?
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Customer Behavior Analysis

Questions

1. Who books furthest in advance?
 2. Who stays longest?
 3. Which guests make special requests most often?
 4. Which guest group is most profitable?
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Part 11: Data Transformation

Apply:

- StandardScaler
- MinMaxScaler
- RobustScaler

Compare:

- Mean
- Standard Deviation
- Distribution

Questions

1. Which scaler handled outliers best?
 2. Which scaler would you use before KNN?
 3. Which scaler would you use before Neural Networks?
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Part 12: Skewness Analysis

Calculate skewness for all numerical features.

Apply:

- Log Transformation
- Square Root Transformation

Compare:

- Before
- After

Questions

1. Which feature was most skewed?
 2. Which transformation improved normality most?
 3. Why is skewness harmful for certain ML algorithms?
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Part 13: Data Engineering Quality Checks

Create rules for:

Null Validation

Examples:

- ADR should not be null.
- Hotel should not be null.

Negative Values

Examples:

- Lead Time < 0
- ADR < 0
- Adults < 0

Business Rules

Examples:

- Adults + Children + Babies > 0
- ADR > 0
- Stay Duration > 0

Data Type Rules

Examples:

- Hotel → String
- ADR → Numeric
- Arrival Date → Datetime

Questions

1. Which rules failed?
 2. Which records violate business constraints?
 3. How would you automate these checks?
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Part 14: Machine Learning Readiness Assessment

Prepare a complete ML Readiness Report.

Include:

Columns to Drop

Explain why.

Columns to Encode

Choose:

- Label Encoding
- One Hot Encoding

Columns to Scale

Explain why.

Highly Correlated Features

Recommend whether to keep or remove.

Feature Selection

Select top features for cancellation prediction.

Questions

1. Which features are most important for predicting cancellations?
2. Which features may cause data leakage?
3. Which features should be excluded?

Part 15: Executive Business Insights Report

Generate at least **15 business insights** supported by visualizations.

Important Insight Questions

1. Why are customers cancelling bookings?
 2. Which hotel performs better?
 3. Which market segment generates maximum revenue?
 4. Which countries generate most bookings?
 5. Which booking channel performs best?
 6. What is the peak booking season?
 7. Which customer type is most valuable?
 8. Which deposit type reduces cancellations?
 9. Does booking early increase cancellation probability?
 10. Which guest type stays longest?
 11. Which guests make most special requests?
 12. Which months require maximum staffing?
 13. Which segment should receive marketing focus?
 14. Which factors most influence cancellations?
 15. How can the hotel increase occupancy and revenue?
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Bonus Challenge (30 Marks)

Create a reusable OOP project:

```
class HotelBookingAnalyzer:
    pass
```

The class should automatically:

- Generate Dataset Summary
- Detect Missing Values
- Detect Duplicates
- Detect Outliers
- Generate EDA Report
- Generate Business Insights
- Generate Revenue Analysis
- Generate Cancellation Analysis
- Generate Data Quality Report
- Generate ML Readiness Report
- Export Reports as PDF

Submission Requirements

1. Jupyter Notebook (.ipynb)
2. Cleaned Dataset (.csv)
3. Data Quality Report (.pdf)
4. Business Insights Report (.pdf)
5. Machine Learning Readiness Report (.pdf)